Lesson 10.1 - Hypothesis Testing - Null and Alternative Hypotheses

> **Video:** Hypothesis Testing - Null and Alternative Hypotheses by The Organic Chemistry Tutor **Chapter:** 10 - Hypothesis Testing - 1 Sample

The two hypotheses

	Null hypothesis H_0	Alternative hypothesis H_a (or H_1)
What it is	The **status quo**: the currently accepted value or claim	The **challenger**: what the researcher is trying to show
Sign	Contains the **equality**: $=, \leq, \geq$	**Strict** inequality: $\neq, <, >$

The researcher tries to gather enough evidence to **reject** H_0 in favor of H_a.

> **Careful wording:** if the evidence isn't strong enough, we **fail to reject** H_0. That doesn't prove H_0 is true, or even that it is "likely" true. It only means the data didn't rule it out.

Steps

1. **Mean or proportion?**
   - "average", "mean" → population mean μ
   - "percent", "proportion", "fraction" → population proportion p, written as a decimal (80% → 0.80)
2. **Find the accepted value** (the status quo) and write H_0 with the equality.
3. **Find what the researcher suspects** and write H_a with the strict inequality. Together, H_0 and H_a cover every possibility.

Wording	H_0	H_a
"is different from", "is not"	$=$	$\neq$
"at least", "no less than"	$\geq$	$<$
"at most", "no more than"	$\leq$	$>$

Why the signs go this way

1. H_0 gets the equality because the math needs one exact number. A test starts by assuming H_0 is true, then asks: "If that were true, how surprising is my sample?" To measure "how surprising" you need one specific value to plug into the formula.

- $\mu = 10$ gives you 10, so you can calculate.
- $\mu < 10$ doesn't. Is it 9.9? 7? 2? You can't compute anything from that.

So the exact number (the equality) always lives in H_0. Even when H_0 is $\geq$ or $\leq$, the calculation uses the boundary value, the "=" part.

2. H_0 gets the benefit of the doubt. Think of a courtroom:

- H_0 = **innocent** (the default, believed until proven otherwise)
- H_a = **guilty** (what the prosecutor has to prove)

The researcher is the prosecutor. The burden of proof is on H_a, so H_a gets the strict claim ($\neq$, $<$, $>$).

3. Together they cover every possibility, with no overlap. That's why they come in fixed pairs:

H_0	H_a	Together
$=$	$\neq$	equal or not equal → everything
$\geq$	$<$	at least, or below → everything
$\leq$	$>$	at most, or above → everything

Example. A company says its battery lasts **at least 10 hours**. You suspect it's less.

$$H_0 : \mu \geq 10 \qquad H_a : \mu < 10$$

H_0 is the company's claim (status quo, includes the "="). H_a is your suspicion (strict).

What "fail to reject" means

"Fail to reject H_0" = "not enough evidence to say H_0 is wrong." It does NOT mean "H_0 is right."

- **Courtroom:** the jury says "not guilty", never "innocent". Maybe the person really is innocent, or maybe the prosecutor just didn't have enough evidence. The verdict only says the case wasn't proven.
- **Lost keys:** you search your room for 30 seconds and don't find your keys. That doesn't prove they aren't there. You just didn't find them. Maybe you didn't look long enough, which in statistics is like having too small a sample.
- **Battery:** you test 5 batteries and get an average of 9.8 hours. That's a bit below 10, but with only 5 batteries it could easily be random luck. So you fail to reject H_0. You haven't proven the batteries last 10+ hours. You just can't call the company a liar yet.

Decision	Meaning
Reject H_0	Strong evidence. H_0 looks wrong, so we side with H_a.
Fail to reject H_0	Not enough evidence. We can't say H_0 is wrong. (Never say "accept H_0".)

Only two decisions: reject or fail to reject

There are only **two** possible decisions about H_0. "Accept H_0" is **not** a third option; it's just a wrong way of saying "fail to reject". A test can only collect evidence **against** H_0, never evidence **for** it.

The courtroom has the same two verdicts:

- **Guilty** = reject H_0
- **Not guilty** = fail to reject H_0

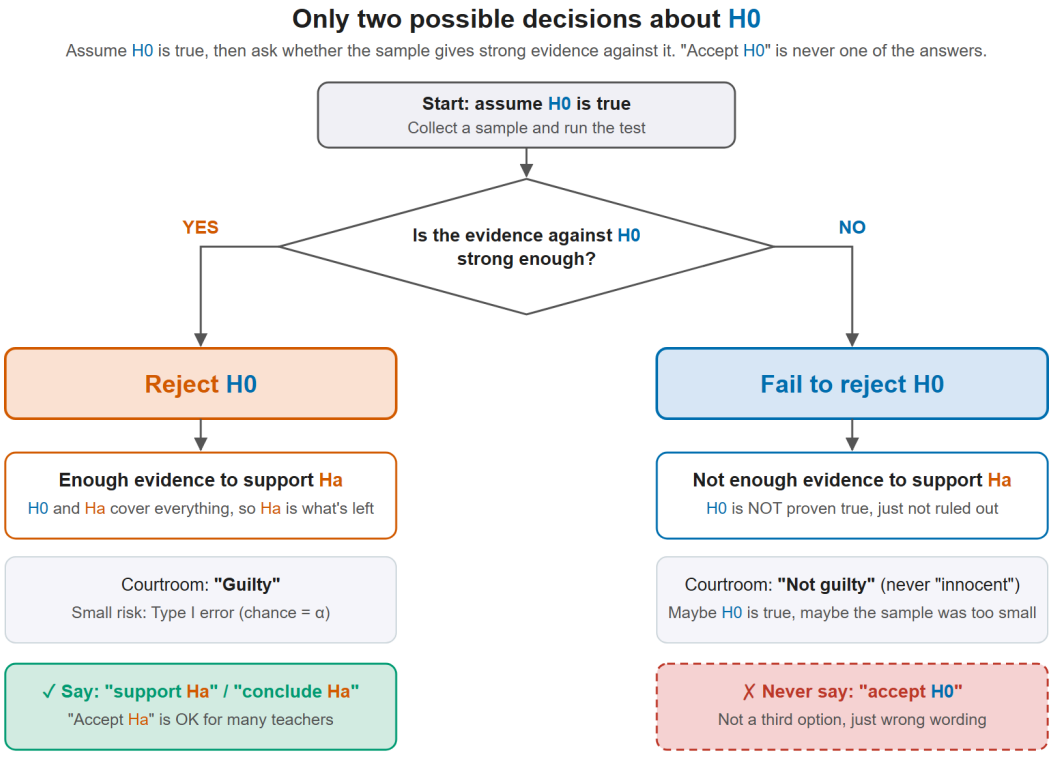
There's no "innocent" verdict, just as there's no "accept H_0".

Reject H_0 means you support H_a. H_0 and H_a cover every possibility with no overlap. If you have strong evidence that H_0 is wrong, the only option left is H_a. It doesn't work the other way: failing to reject H_0 says

nothing in favor of H_0.

Decision	What you can say about H_a
Reject H_0	Enough evidence to **support** H_a
Fail to reject H_0	Not enough evidence to support H_a (and H_0 is not proven true)

> **Small caution:** "support H_a" or "conclude H_a" is safer wording than "accept H_a", because rejecting H_0 isn't 100% proof. There's a small chance you rejected H_0 when it was actually true. That's a **Type I error**, and its chance is α (for example 5%). See Lesson 10.2.



Exam-friendly wording:

- ✅ "Reject H_0. There is sufficient evidence to support the claim that $\mu < 10$."
- ✅ "Fail to reject H_0. There is not enough evidence to support the claim that $\mu < 10$."
- ❌ "Accept H_0." (Some older textbooks use it, but most teachers mark it wrong.)

Examples

1. Calculator mass. Company XYZ's calculators average **450 g**. An engineer thinks the average is different and weighs 50 calculators.

$$H_0 : \mu = 450 \qquad H_a : \mu \neq 450$$

("Average" means a mean.)

2. Graduation rate. Teachers believe **at least 80%** of students graduate. A student disagrees.

$$H_0 : p \geq 0.80 \qquad H_a : p < 0.80$$

("Percent" means a proportion. "At least 80%" means 80% or more.)

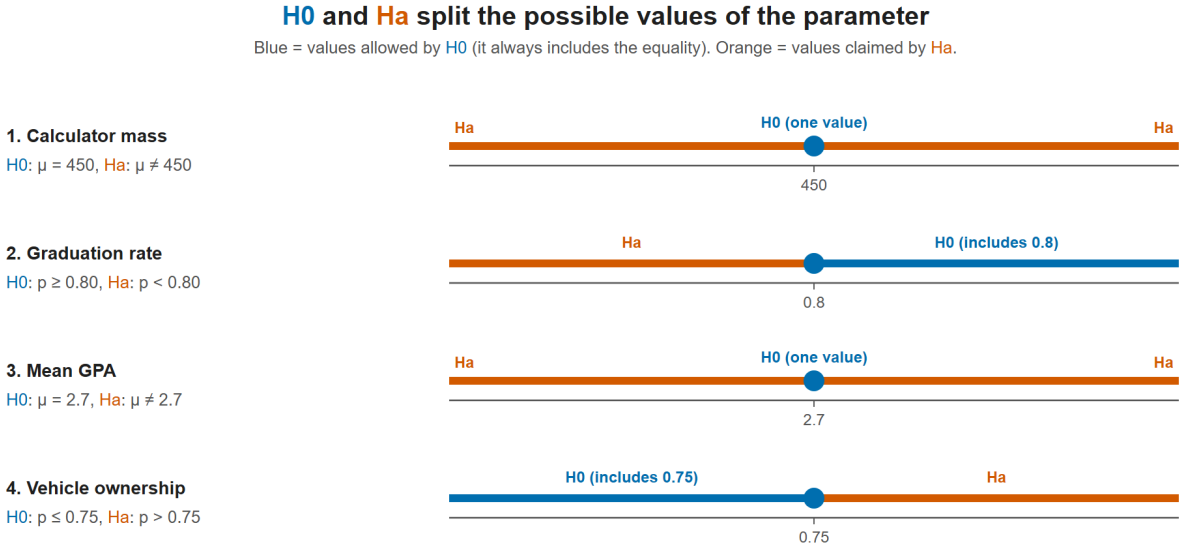
3. GPA. A teacher wants to test whether the mean GPA is **different from 2.7**. Here it is easier to start with H_a, the teacher's claim.

$$H_0 : \mu = 2.7 \qquad H_a : \mu \neq 2.7$$

4. Vehicle ownership. The share of residents who own a vehicle is **no more than 75%**. A researcher disagrees and surveys 100 residents.

$$H_0 : p \leq 0.75 \qquad H_a : p > 0.75$$

("No more than" means ≤. Surveying 100 residents gives the sample size, which is not part of either hypothesis.)



H_0 and H_a always cover every possible value between them. H_0 holds the boundary value (the equality). With "≠", H_0 is a single point and H_a is everything else.

Key takeaways

- H_0 = status quo with the equality ($=$, $\leq$, $\geq$). H_a = the claim being tested, with a strict sign ($\neq$, $<$, $>$).
- Hypotheses are about **population parameters** (μ or p), never about sample values like $\bar{x}$ or $\hat{p}$.
- Translate the wording carefully: "at least" → $\geq$, "no more than" → $\leq$, "different" → $\neq$.
- H_0 holds the equality because the test needs one exact value to calculate with, and H_0 gets the benefit of the doubt (like "innocent until proven guilty").
- "Fail to reject H_0" means "not proven wrong", not "proven right". Never say "accept H_0".
- Only two decisions exist: **reject** H_0 (→ support H_a) or **fail to reject** H_0 (→ not enough evidence for H_a).

Lesson 10.2 - How to Identify Type I and Type II Errors

> **Video:** How To Identify Type I and Type II Errors In Statistics by The Organic Chemistry Tutor **Chapter:** 10 - Hypothesis Testing - 1 Sample

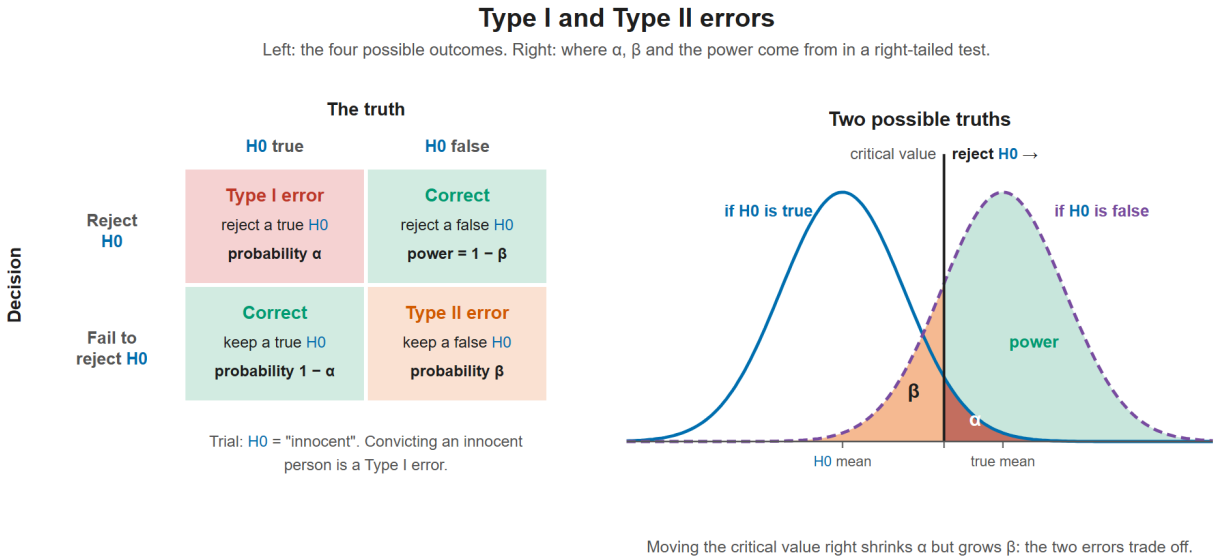
Definitions

Error	What happens	Probability
Type I	**Reject** H_0 when H_0 is actually **true**	α (the significance level)
Type II	**Fail to reject** H_0 when H_0 is actually **false**	β
(correct) Power	Reject H_0 when it is false	$1-\beta$, the **power of the test**

Decision table

	H_0 is **true**	H_0 is **false**
Reject H_0	❌ **Type I error** (α)	✅ Correct (power $= 1-\beta$)
Fail to reject H_0	✅ Correct	❌ **Type II error** (β)

We can't control whether H_0 is true. We only control the decision. (The video sometimes says "accept H_0"; the standard wording is "fail to reject H_0".)



Left: the four combinations of decision and truth. Right: if H_0 is true, the red tail beyond the critical value (α) is the chance of a false alarm. If H_0 is false, the orange area left of the critical value (β) is the chance of missing the effect, and the green area is the power.

How to identify the error type

1. Mark each scenario as a **good** decision or a **bad** decision (an error).
2. For each error, ask whether the facts make H_0 **true or false**:

- H_0 true but rejected → **Type I**
- H_0 false but not rejected → **Type II**

Example 1 - is John's car safe?

H_0: *John's used car is safe to drive.*

Choice	Situation	Good or bad?	Error
A	Thinks the car is safe, but it is **not** safe	Bad	H_0 false, not rejected → **Type II**
B	Thinks the car is safe, and it **is** safe	Good	—
C	Thinks it may not be safe, and it is **not** safe	Good	—
D	Thinks it may not be safe, but it **is** safe	Bad	H_0 true, rejected → **Type I**

Which error is worse? The Type II error (A): he drives an unsafe car and could be badly hurt. The Type I error (D) only means he stays home unnecessarily.

Example 2 - a criminal trial

H_0: *The defendant is innocent* (presumed innocent).

Choice	Situation	Good or bad?	Error
A	Jury says **guilty**, but he is **innocent**	Bad	H_0 true, rejected → **Type I**
B	Jury says guilty, and he is **not** innocent	Good	—
C	Jury says **not guilty**, but he is **not** innocent	Bad	H_0 false, not rejected → **Type II**
D	Jury says not guilty, and he **is** innocent	Good	—

Which error is worse? Usually the Type I error: an innocent person goes to prison. Letting a guilty person go free (Type II) is also bad, but courts are designed to avoid convicting the innocent. This is partly a judgment call.

Key takeaways

- Type I = **false alarm**: rejecting a true H_0, probability α.
- Type II = **missed detection**: failing to reject a false H_0, probability β.
- Power = $1 - \beta$, the probability of correctly rejecting a false H_0.
- Which error is "worse" depends on the real-world consequences in context.

Lesson 10.3 - One Tailed and Two Tailed Tests

> **Video:** One Tailed and Two Tailed Tests, Critical Values, & Significance Level by The Organic Chemistry Tutor **Chapter:** 10 - Hypothesis Testing - 1 Sample

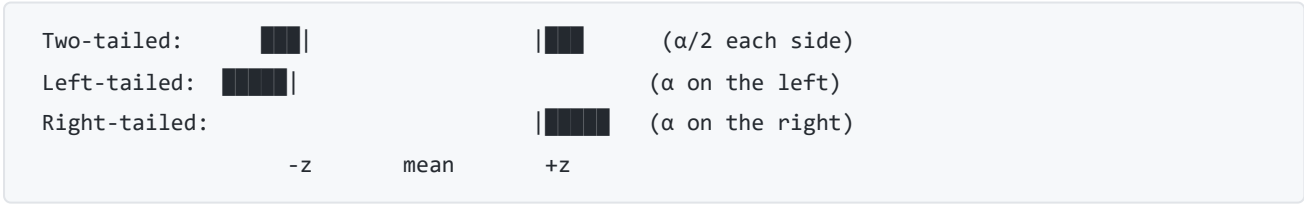
Vocabulary

Term	Meaning
Rejection region	The shaded tail area(s). If the test statistic lands here, reject H_0
Fail-to-reject region	The unshaded middle. If the statistic lands here, keep H_0
Critical value(s)	The z (or t) value(s) separating the two regions
Confidence level c	The area of the fail-to-reject region
Significance level α	The total area of the rejection region: $\alpha = 1 - c$
Calculated value z_c	The **test statistic** computed from the sample (see Lesson 10.4)

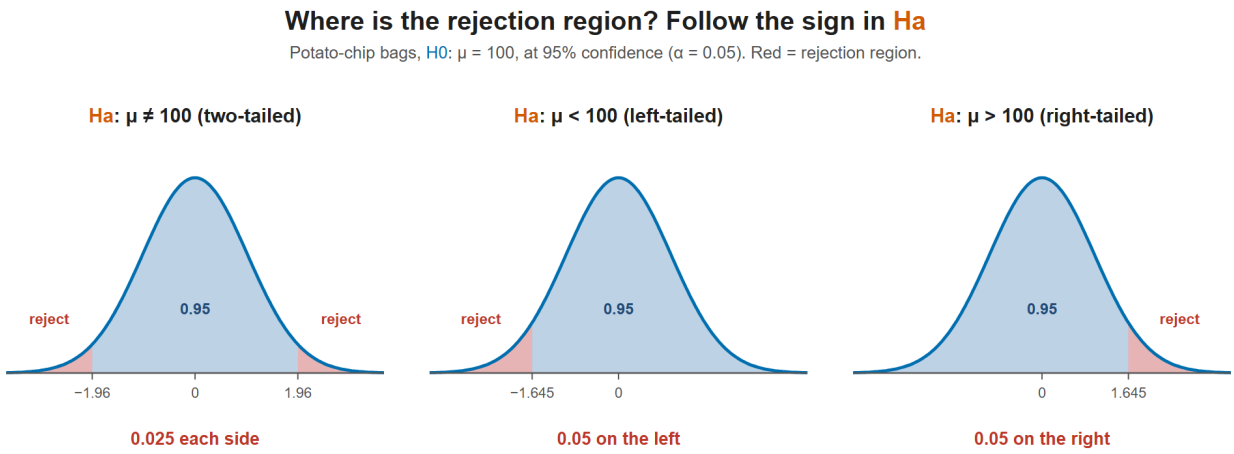
Decision: if z_c falls in the rejection region (beyond the critical value), **reject** H_0. Otherwise, **fail to reject** H_0.

Which test? Read the sign in H_a

H_a	Test	Rejection region	At 95% ($\alpha = 0.05$)
$\mu \neq \mu_0$	**Two-tailed**	Both tails, $\alpha/2$ each	0.025 in each tail, critical $z = \pm 1.96$
$\mu < \mu_0$	**Left-tailed**	Left tail only, area α	0.05 on the left, critical $z = -1.645$
$\mu > \mu_0$	**Right-tailed**	Right tail only, area α	0.05 on the right, critical $z = +1.645$



The **arrow** in H_a points to the tail: $<$ → left, $>$ → right, $\neq$ → both.



The sign in H_a decides where α goes: split between both tails for ≠, all in the left tail for <, and all in the right tail for >.

Example - potato-chip bags

A company says its chip bags average **100 g**.

- An employee thinks the mean is **not** 100 g: $H_0 : \mu = 100$, $H_a : \mu \neq 100$ → **two-tailed**. At 95% confidence, $\alpha = 0.05$ is split into 0.025 per tail, and 0.95 is in the middle.
- If instead $H_a : \mu < 100$ → **left-tailed**, with all 0.05 in the left tail.
- If instead $H_a : \mu > 100$ → **right-tailed**, with all 0.05 in the right tail.

Key takeaways

- $\alpha = 1 - c$ is the significance level, the size of the rejection region.
- $\neq$ → two tails (split α). $<$ → left tail. $>$ → right tail.
- Compare the calculated statistic with the critical value(s) to decide.

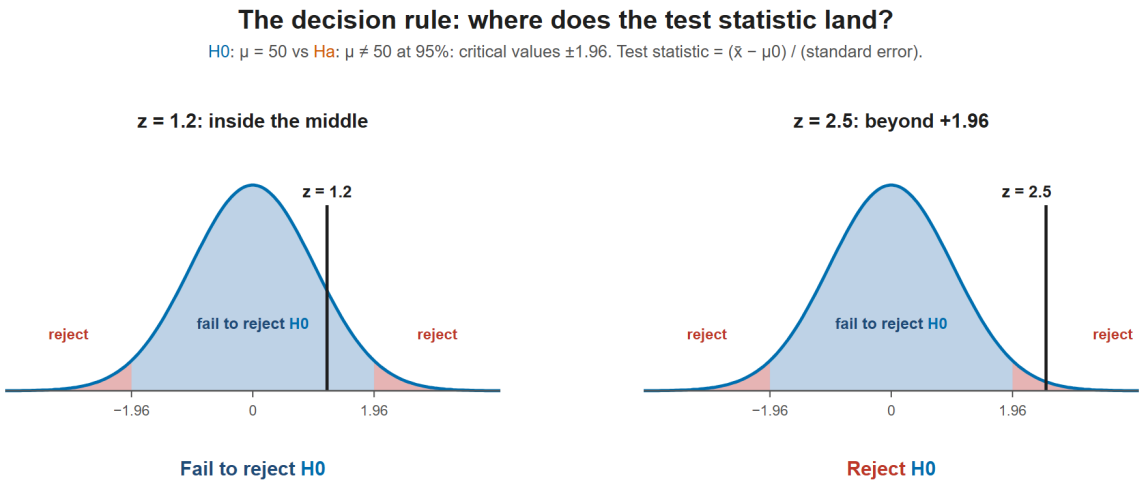
Lesson 10.4 - Test Statistic for Means and Population Proportions

> **Video:** Test Statistic For Means and Population Proportions by The Organic Chemistry Tutor **Chapter:** 10 - Hypothesis Testing - 1 Sample

The decision rule

1. Find the **critical value(s)** from the significance level and the type of test. For a two-tailed test at 95%, ± 1.96.
2. Compute the **test statistic** z_c (or t) from the sample.
3. Statistic in the rejection region → **reject** H_0. Statistic in the fail-to-reject region → **fail to reject** H_0.

Example: $H_0 : \mu = 50$ vs $H_a : \mu \neq 50$ at 95% has critical values ± 1.96. A calculated z_c beyond 1.96 (or below −1.96) means reject.



Compute the test statistic, then see which region it lands in. Between −1.96 and 1.96, fail to reject H_0. Beyond either critical value, reject H_0.

Test statistic for a mean

Sample size	σ known?	Distribution	Test statistic
$n < 30$	**No** (only s)	**t**, df $= n - 1$	$t = \dfrac{\bar{x} - \mu}{s/\sqrt{n}}$
$n < 30$	Yes	z	$z = \dfrac{\bar{x} - \mu}{\sigma/\sqrt{n}}$
$n \geq 30$	No (use s)	z (approximately normal)	$z = \dfrac{\bar{x} - \mu}{s/\sqrt{n}}$
$n \geq 30$	Yes	z	$z = \dfrac{\bar{x} - \mu}{\sigma/\sqrt{n}}$

Here μ is the value claimed in H_0.

> Many textbooks use the t-distribution **whenever σ is unknown**, whatever the sample size. For $n \geq 30$, t and z give nearly the same answer, so the video's rule is a common shortcut.

Test statistic for a proportion

$$z = \frac{\hat{p} - p}{\sqrt{\dfrac{pq}{n}}}, \qquad q = 1 - p$$

- $\hat{p}$ = sample proportion (x/n)
- p = the proportion claimed in H_0 (sometimes written p_0). Use this hypothesized value under the square root, not $\hat{p}$.

Key takeaways

- Test statistic = (sample value − hypothesized value) ÷ standard error.
- Mean: use t with s when $n < 30$ and σ is unknown. Otherwise use z.
- Proportion: $z = (\hat{p} - p)/\sqrt{pq/n}$, with p from H_0.
- Compare with the critical value(s) to reject or fail to reject H_0.

Lesson 10.5 - Hypothesis Testing Problems - Z Test and T Statistics

> **Video:** Hypothesis Testing Problems - Z Test & T Statistics - One & Two Tailed Tests by The Organic Chemistry Tutor **Chapter:** 10 - Hypothesis Testing - 1 Sample

Procedure (critical-value method)

1. State H_0 (status quo, with the equality) and H_a (the claim being tested).
2. Decide on the tails from H_a: $\neq$ → two-tailed, $<$ → left, $>$ → right.
3. Find the **critical value** from α (z-table, or t-table with df $= n - 1$).
4. Compute the **test statistic**: $\dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$, treated as z if $n \geq 30$ or as t if $n < 30$ and σ is unknown.
5. If the statistic is in the rejection region, **reject** H_0. Otherwise, **fail to reject** H_0.

Example 1 - two-tailed z-test (a bottling machine)

A machine is supposed to dispense **80 mL**. An employee thinks the mean is not 80 mL. A sample of **40** bottles has $\bar{x} = 78$ mL and $s = 2.5$ mL.

(a) Hypotheses

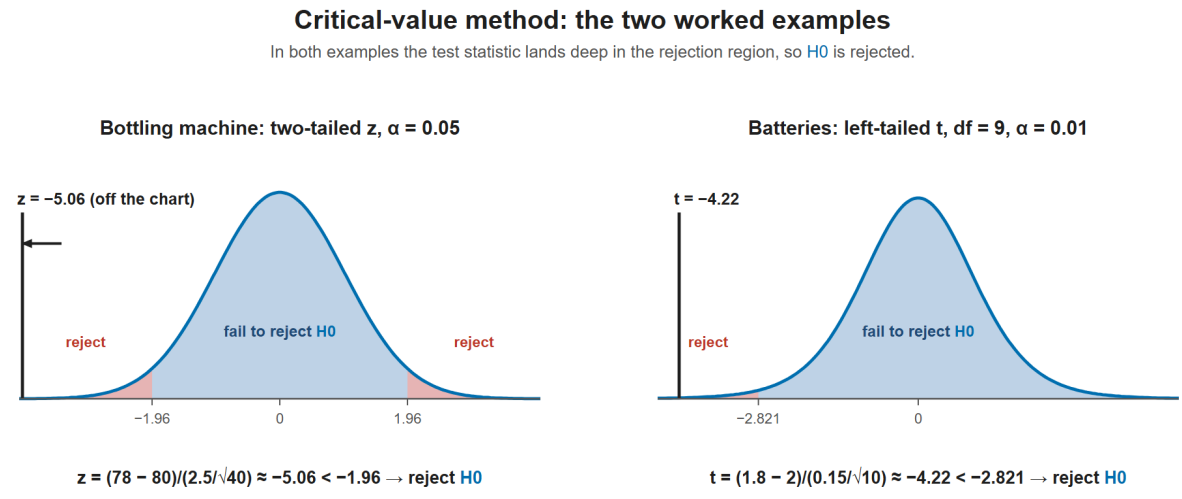
$$H_0 : \mu = 80 \qquad H_a : \mu \neq 80$$

(b) Is there evidence at 95% confidence that the machine is off?

- H_a uses $\neq$, so this is **two-tailed**: 0.025 in each tail, critical values $z = \pm 1.96$.
- $n = 40 \geq 30$, so use z (with s in place of σ).

$$z = \frac{78 - 80}{2.5/\sqrt{40}} = \frac{-2}{0.39528} \approx -5.06$$

$-5.06 < -1.96$, so the statistic is far into the left rejection region. **Reject H_0.** At the 5% significance level there is strong evidence that the machine does not dispense 80 mL on average.



Both worked examples. The bottling statistic (−5.06) is so extreme it lies off the chart. The battery test uses a t-distribution with 9 degrees of freedom, so its left critical value is −2.821, not a z value.

Example 2 - left-tailed t-test (car batteries)

The company says its batteries last **2 or more years** on average. An engineer thinks it's less. A sample of **10** batteries has $\bar{x} = 1.8$ years and $s = 0.15$ years.

(a) Hypotheses

$$H_0 : \mu \geq 2 \qquad H_a : \mu < 2$$

(b) Test at 99% confidence

- H_a uses $<$, so this is **left-tailed**, with $\alpha = 0.01$ entirely in the left tail.
- $n = 10 < 30$ and σ is unknown, so use the **t-distribution** with df $= 10 - 1 = 9$.
- One-tail t-table, df = 9, $\alpha = 0.01$: $t = 2.821$, so the critical value is -2.821 (left side).

$$t = \frac{1.8 - 2}{0.15/\sqrt{10}} = \frac{-0.2}{0.047434} \approx -4.22$$

$-4.22 < -2.821$, which is in the rejection region. **Reject H_0.** At the 1% significance level there is enough evidence that the mean battery life is less than 2 years.

> **Wording:** the video sometimes says "accept the null hypothesis". The standard phrasing is **"fail to reject H_0"**. Not rejecting doesn't prove H_0 is true.

Key takeaways

- H_a decides the tails. n and whether σ is known decide z or t.
- Test statistic = (sample mean − hypothesized mean) ÷ (SD / √n).
- If the statistic is beyond the critical value, reject H_0. Otherwise, fail to reject H_0.
- A 95% (99%) "confidence level" in a test corresponds to $\alpha = 0.05$ (0.01).

Lesson 10.6 - P-Value Method for Hypothesis Testing

> **Video:** P-Value Method For Hypothesis Testing by The Organic Chemistry Tutor **Chapter:** 10 - Hypothesis Testing - 1 Sample

Two ways to make the decision

	Traditional (critical-value) method	**P-value method**
Compare	calculated z **vs** critical z	**p-value vs** α
Reject H_0 when	z is in the rejection region	p-value $< \alpha$
Fail to reject when	z is in the fail-to-reject region	p-value $\geq \alpha$

Significance level: $\alpha = 1 - \text{confidence level}$.

p-value = the area in the tail(s) beyond the calculated test statistic:

Test	p-value
Left-tailed	area to the **left** of z
Right-tailed	area to the **right** of z
Two-tailed	**2 ×** the area in the tail beyond $\lvert z \rvert$

Example 1 - two-tailed (town weights)

Town XYZ's mean weight is claimed to be **168 lb**. A nutritionist thinks it is different. A sample of **36** people has $\bar{x} = 169.5$ lb and $s = 3.9$ lb.

(a)

$$H_0 : \mu = 168 \qquad H_a : \mu \neq 168$$

(b) Test at 95% confidence ($\alpha = 0.05$), two-tailed, $n \geq 30$ so use z:

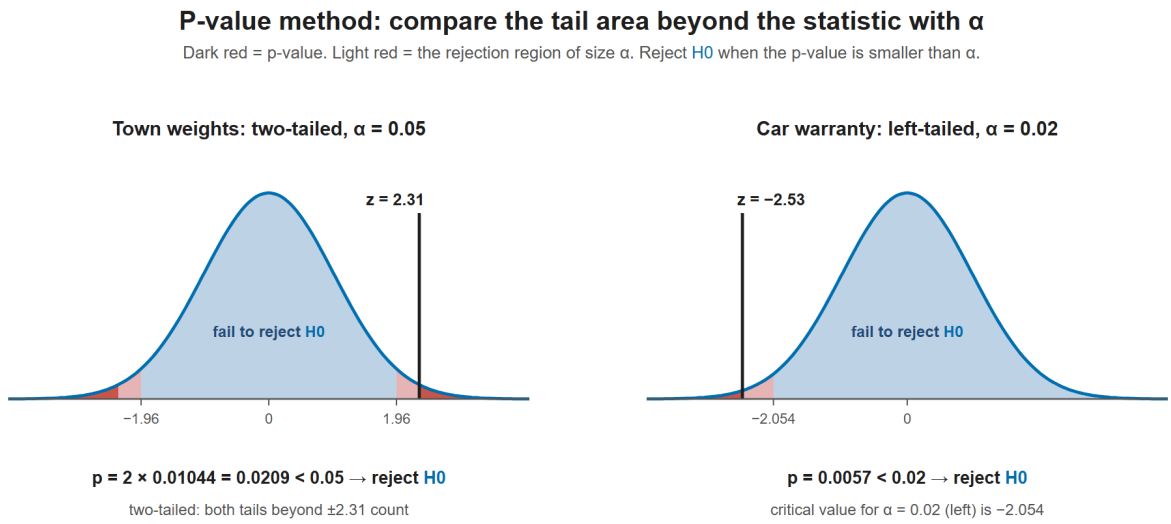
$$z = \frac{169.5 - 168}{3.9/\sqrt{36}} = \frac{1.5}{0.65} \approx 2.31$$

Traditional method: the critical values are ± 1.96 (left area 0.975). $2.31 > 1.96$, so it is in the rejection region → **reject** H_0.

P-value method:

- The table gives a left area of 0.98956 for $z = 2.31$.
- Right tail: $1 - 0.98956 = 0.01044$.
- Two-tailed p-value: $2 \times 0.01044 = 0.02088$.

$0.02088 < 0.05$ → **reject** H_0. At the 5% level there is enough evidence that the mean weight is not 168 lb.



The p-value (dark red) is the tail area beyond the test statistic. For a two-tailed test, both tails count. In both examples the p-value is smaller than α (the light red region), so H_0 is rejected.

Example 2 - left-tailed (car warranty)

A factory gives a **5-year** warranty on the engine and transmission. An engineer believes failures happen **in less than 5 years**. A sample of **40** cars has $\bar{x} = 4.8$ years and $s = 0.5$ years.

(a)

$$H_0 : \mu \geq 5 \qquad H_a : \mu < 5$$

(b) Is there evidence at the 2% significance level ($\alpha = 0.02$, 98% confidence) to revise the warranty?

This is left-tailed (shade the left only), and $n = 40 \geq 30$ so use z:

$$z = \frac{4.8 - 5}{0.5/\sqrt{40}} = \frac{-0.2}{0.07906} \approx -2.53$$

The p-value is the area to the left of −2.53 = **0.0057**.

$0.0057 < 0.02$ → **reject** H_0. There is enough evidence that the mean time to failure is under 5 years, so the warranty should be revised.

Key takeaways

- p-value = the probability, assuming H_0 is true, of a result at least as extreme as the one observed.
- **Small p-value (< α) → reject H_0.** Large p-value (≥ α) → fail to reject.
- For a two-tailed test, **double** the one-tail area.
- The p-value method and the critical-value method always give the same decision.

Lesson 10.7 - Hypothesis Testing - Solving Problems with Proportions

> **Video:** Hypothesis Testing - Solving Problems With Proportions by The Organic Chemistry Tutor **Chapter:** 10 - Hypothesis Testing - 1 Sample

Formulas

Quantity	Formula
Sample proportion	$\hat{p} = \dfrac{x}{n}$
Hypothesized proportion (from H_0)	p_0, with $q_0 = 1 - p_0$
Test statistic	$z = \dfrac{\hat{p} - p_0}{\sqrt{\dfrac{p_0 q_0}{n}}}$

"Percent" in the problem means you are testing a **proportion** p, not a mean μ.

Example 1 - two-tailed (cell-phone ownership)

A tech company says **70%** of residents own a cell phone. A marketing manager thinks it is different. In a survey of **200**, **130** said yes.

(a)

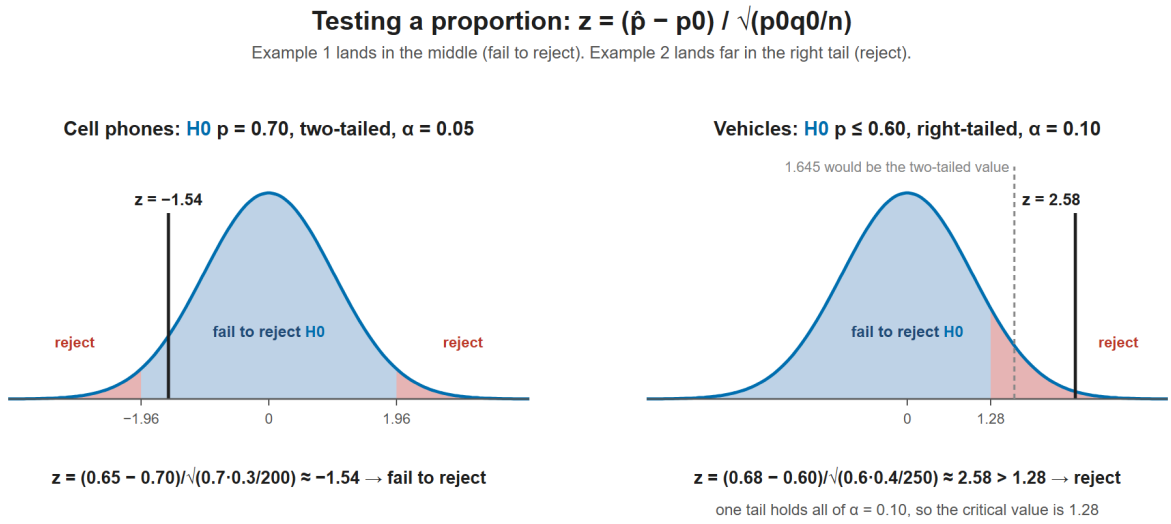
$$H_0 : p = 0.70 \qquad H_a : p \neq 0.70$$

(b) At 95% confidence, is there enough evidence to reject H_0?

- $\hat{p} = \dfrac{130}{200} = 0.65,\ \ p_0 = 0.70,\ \ q_0 = 0.30$
- $\neq$, so two-tailed: 0.025 per tail, critical values ± 1.96

$$z = \frac{0.65 - 0.70}{\sqrt{\dfrac{0.70 \times 0.30}{200}}} = \frac{-0.05}{\sqrt{0.00105}} = \frac{-0.05}{0.0324} \approx -1.54$$

$-1.96 < -1.54 < 1.96$, which is in the fail-to-reject region. **Fail to reject H_0.** At the 5% level there is not enough evidence that the proportion differs from 70%.



Example 1 stays in the fail-to-reject region. Example 2 is right-tailed, so all 10% of α sits in one tail and the critical value is 1.28. The dashed 1.645 is the value you would get by wrongly splitting α between two tails.

Example 2 - right-tailed (vehicle ownership)

A car company says **60% or less** of residents own a vehicle. A sales manager disagrees. In a survey of **250**, **170** said yes.

(a)

$$H_0 : p \leq 0.60 \qquad H_a : p > 0.60$$

(b) At the 10% significance level, is there enough evidence?

- $\hat{p} = \dfrac{170}{250} = 0.68,\ p_0 = 0.60,\ q_0 = 0.40$
- $>$, so this is a **right-tailed** test with **all** of $\alpha = 0.10$ in the right tail.
- Critical value: the left area is $1 - 0.10 = 0.90$, which gives $z = 1.28$.

$$z = \frac{0.68 - 0.60}{\sqrt{\dfrac{0.60 \times 0.40}{250}}} = \frac{0.08}{\sqrt{0.00096}} = \frac{0.08}{0.03098} \approx 2.58$$

$2.58 > 1.28$, which is in the rejection region. **Reject H_0.** At the 10% level there is enough evidence that more than 60% of residents own a vehicle.

> **Correction to the video:** the video splits $\alpha = 0.10$ into 5% in each tail and uses a critical value of 1.645. That split is only for **two-tailed** tests. For this **right-tailed** test the whole 10% is in one tail, so the critical value is **1.28**. Here the conclusion is the same, because 2.58 exceeds both.

Key takeaways

- Proportion test: $z = (\hat{p} - p_0)/\sqrt{p_0 q_0 / n}$, using the **null** value p_0 in the standard error.
- Two-tailed: split α between the tails. One-tailed: put all of α in one tail.

- Critical z values: two-tailed 10% → ±1.645, 5% → ±1.96. One-tailed 10% → 1.28, 5% → 1.645.